

14/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-4

Time : 180 Mins.

Answers

1	(4)	37	(3)	73	(2)	109	(4)	145	(4)
2	(2)	38	(4)	74	(3)	110	(1)	146	(1)
3	(1)	39	(2)	75	(1)	111	(4)	147	(2)
4	(1)	40	(4)	76	(4)	112	(2)	148	(3)
5	(3)	41	(3)	77	(4)	113	(4)	149	(1)
6	(3)	42	(1)	78	(4)	114	(2)	150	(3)
7	(2)	43	(2)	79	(3)	115	(4)	151	(4)
8	(2)	44	(2)	80	(4)	116	(4)	152	(3)
9	(3)	45	(3)	81	(3)	117	(3)	153	(3)
10	(1)	46	(4)	82	(4)	118	(3)	154	(2)
11	(4)	47	(2)	83	(3)	119	(2)	155	(3)
12	(1)	48	(1)	84	(4)	120	(3)	156	(4)
13	(3)	49	(4)	85	(3)	121	(1)	157	(3)
14	(3)	50	(1)	86	(2)	122	(2)	158	(4)
15	(3)	51	(3)	87	(3)	123	(2)	159	(2)
16	(2)	52	(3)	88	(1)	124	(2)	160	(4)
17	(1)	53	(3)	89	(4)	125	(4)	161	(1)
18	(2)	54	(3)	90	(3)	126	(2)	162	(4)
19	(4)	55	(4)	91	(2)	127	(4)	163	(2)
20	(2)	56	(3)	92	(4)	128	(3)	164	(3)
21	(3)	57	(1)	93	(4)	129	(2)	165	(3)
22	(3)	58	(2)	94	(3)	130	(3)	166	(3)
23	(2)	59	(3)	95	(2)	131	(4)	167	(4)
24	(3)	60	(1)	96	(4)	132	(4)	168	(4)
25	(3)	61	(1)	97	(2)	133	(3)	169	(3)
26	(1)	62	(4)	98	(4)	134	(3)	170	(2)
27	(2)	63	(4)	99	(2)	135	(3)	171	(4)
28	(2)	64	(3)	100	(2)	136	(4)	172	(4)
29	(1)	65	(1)	101	(4)	137	(2)	173	(3)
30	(3)	66	(4)	102	(4)	138	(4)	174	(3)
31	(1)	67	(4)	103	(2)	139	(1)	175	(2)
32	(3)	68	(2)	104	(2)	140	(4)	176	(3)
33	(1)	69	(2)	105	(2)	141	(1)	177	(1)
34	(2)	70	(4)	106	(1)	142	(4)	178	(4)
35	(1)	71	(1)	107	(3)	143	(2)	179	(3)
36	(3)	72	(3)	108	(2)	144	(4)	180	(4)

Hints and Solutions

PHYSICS

(1) Answer : (4)

Solution:

Density of nucleus remain same for all nucleus.

(2) Answer : (2)

Solution:

Energy of ground state of hydrogen atom is -13.6 eV.

$$\begin{aligned}\text{Therefore, } E_3 &= \frac{-13.6}{n^2} \\ &= \frac{13.6}{9} = -1.51 \text{ eV}\end{aligned}$$

(3) Answer : (1)

Solution:

When both switch will be closed then only we get output.

$\therefore$ AND gate

(4) Answer : (1)

Solution:

In time t_0 , its amplitude becomes half

$\therefore$ Its half life = t_0

$$\begin{aligned}\Rightarrow \text{Amplitude after time } 3t_0 &= \frac{A}{2^3} \\ &= \frac{40}{2^3} \\ &= \frac{40}{8} \\ &= 5 \text{ cm}\end{aligned}$$

(5) Answer : (3)

Solution:

From $t = 0$ to $t = \frac{\pi}{\omega}$

$$i_{avg} = \frac{2i_0}{\pi}$$

(6) Answer : (3)

Solution:

Wavelength associated with moving particle is given by $\lambda = \frac{h}{p}$ and photons are electrically neutral.

(7) Answer : (2)

Solution:

$$\begin{aligned}\frac{\Delta Q}{\Delta t} &= \frac{(T_1 - T_0) \times K_1 A}{1} = \frac{(T_0 - T_2) \times K_2 A}{1} \\ \Rightarrow (T_1 - T_0) K_1 &= (T_0 - T_2) K_2 \\ T_1 K_1 - T_0 K_1 &= T_0 K_2 - K_2 T_2 \\ \Rightarrow \frac{T_1 K_1 + T_2 K_2}{K_1 + K_2} &= T_0\end{aligned}$$

(8) Answer : (2)

Solution:

In process $a \rightarrow b \Rightarrow$ Volume remains constant

$$\Rightarrow \Delta U = nC_v \Delta T \Rightarrow \Delta U = \frac{1 \times 3R}{2} \times \frac{T_0}{2} = \frac{3RT_0}{4}$$

(9) Answer : (3)

Solution:

The internal energy of an ideal gas is a function of temperature and it is a point function.

(10) Answer : (1)

Solution:

Average translational kinetic energy per molecule of any kind of ideal gas $= \frac{3}{2} K_B T$.

$\Rightarrow$ Ratio = 1 : 1

(11) Answer : (4)

Solution:

Since both the vessels have common height, hence pressure will be same.

Force = Pressure $\times$ Area, since pressure as well as area of base of both the vessels is same, hence force would be same in both cases.

(12) Answer : (1)

Solution:

The iron needle floats due to surface tension.

(13) Answer : (3)

Solution:

$$\vec{x} = 3t^2 \hat{i}$$

$$\vec{v}_x = \frac{d\vec{x}}{dt} = 6t \hat{i}$$

$$\vec{v}_x \text{ (at } t = 2) = 12 \hat{i}$$

$$\text{Similarly } \vec{y} = 5t \hat{j}$$

$$\vec{v}_y = \frac{d\vec{y}}{dt} = 5 \hat{j}$$

Hence, speed of particle at $t = 2$ s

$$\begin{aligned} v &= \sqrt{v_x^2 + v_y^2} \\ &= \sqrt{12^2 + 5^2} \\ &= 13 \text{ m/s} \end{aligned}$$

(14) Answer : (3)

Solution:

$$\text{Angular velocity } \omega = \frac{d\theta}{dt} = 12t - 6t^2$$

Angular acceleration

$$\alpha = \frac{d\omega}{dt} = 12 - 12t = 6 \Rightarrow t = 0.5 \text{ sec}$$

$$\therefore \omega = 12 \times \frac{1}{2} - 6 \times \left(\frac{1}{2}\right)^2 = \frac{9}{2} = 4.5 \text{ rad/s}$$

(15) Answer : (3)

Solution:

$$\text{Minimum deviation } \delta_m = 2i - A = 2 \times 60^\circ - 60^\circ = 60^\circ$$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

(16) Answer : (2)

Solution:

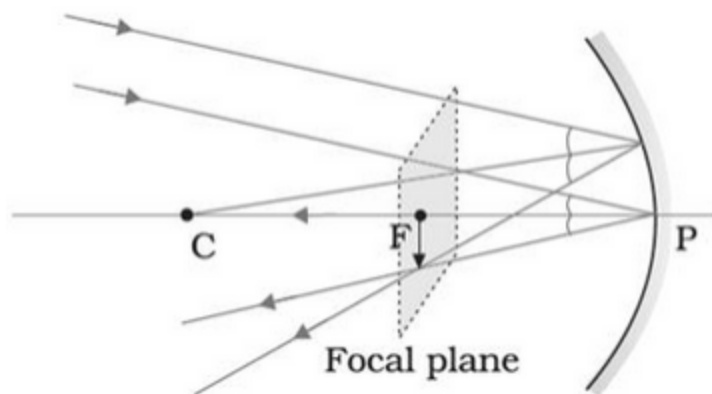
$$\vec{F}_{OA} = \frac{G(2m)(4m)}{2^2} \hat{j} = 2Gm^2 \hat{j}$$

$$\vec{F}_{OB} + \vec{F}_{OC} = \frac{G(m)(4m)}{2^2} \sin 30^\circ \times 2 (-\hat{j}) = -Gm^2 \hat{j}$$

$$\therefore \text{Net force} = \vec{F}_{OA} + \vec{F}_{OB} + \vec{F}_{OC} = Gm^2 \hat{j}$$

(17) Answer : (1)

Solution:



For converging concave mirror, power $P = -\frac{1}{f}$ is positive

(18) Answer : (2)

Solution:

By conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$

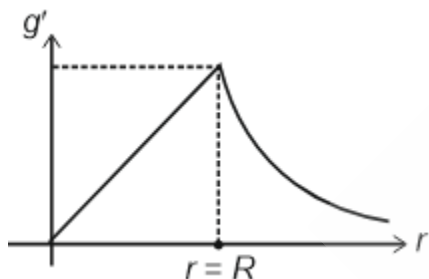
$$\frac{2}{5}MR^2 \times \frac{2\pi}{T_1} = \frac{2}{5}M\left(\frac{R}{\sqrt{3}}\right)^2 \times \frac{2\pi}{T_2}$$

$$\Rightarrow T_2 = \frac{T_1}{3} = \frac{24}{3} = 8 \text{ hours}$$

(19) Answer : (4)

Solution:

$g' = g$ at poles is independent of ω .



i.e., the value of acceleration due to gravity decreases as we move away from surface or move towards centre from surface.

(20) Answer : (2)

Solution:

Young's modulus is a material dependent constant.

(21) Answer : (3)

Solution:

For block to be in contact with piston $a_{\max} = g$.

$$\omega^2 A = g$$

$$A = \frac{g}{\omega^2}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{1}$$

$$A = \frac{10}{4\pi^2} = 0.25 \text{ m}$$

(22) Answer : (3)

Solution:

Travelling wave along y axis is given by

$$x = A \sin(ky - \omega t)$$

(23) Answer : (2)

Solution:

$$\vec{V} = \vec{E} \times \vec{B}$$

$$\vec{V} \perp \vec{E} \perp \vec{B}$$

(24) Answer : (3)

Solution:

$$\tan\theta = \mu$$

$$1 < \mu < \infty$$

$$1 < \tan\theta < \infty$$

$$\tan 45^\circ < \tan\theta < \tan 90^\circ$$

$$45^\circ < \theta < 90^\circ$$

(25) Answer : (3)

Solution:

Gauss law :

$$\phi_{\text{net}} = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{en}}}{\epsilon_0}$$

Hence, net electric flux through a closed surface depends on the charge enclosed by the surface.

(26) Answer : (1)

Solution:

$$\vec{\tau} = \vec{P} \times \vec{E}$$

$$\vec{\tau} = PE \sin\theta = \frac{PE}{2} \dots (i)$$

Work done by external force :

$$W = U_f - U_i = PE (\cos\theta_1 - \cos\theta_2)$$

$$\theta_1 = 30^\circ, \theta_2 = 60^\circ$$

$$\therefore W = PE(\cos 30^\circ - \cos 60^\circ)$$

$$W = PE \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$W = \frac{2\tau}{2} \times (\sqrt{3} - 1) = \tau (\sqrt{3} - 1)$$

(27) Answer : (2)

Solution:

$$\text{Resistance } R = \frac{\rho L}{A}$$

$$\rho = \frac{1}{\sigma} = \frac{RA}{L}$$

$$\sigma = \frac{L}{RA} = \frac{1}{10 \times \pi \left(\frac{10^{-3}}{\sqrt{\pi}} \right)^2}$$

$$\sigma = 10^5$$

$$J = \sigma E = 10^5 \times 10 = 10^6 \text{ A/m}^2$$

(28) Answer : (2)

Solution:

Electric field due to an infinite wire

$$E = \frac{2k\lambda}{r}$$

Net electric field:

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2$$

$$\left| \vec{E}_{\text{net}} \right| = \left| \frac{2K\lambda}{2R} + \frac{2K(-\lambda)}{R} \right| = \left| \frac{-K\lambda}{R} \right| = \frac{K\lambda}{R}$$

(29) Answer : (1)

Solution:

Relation between electric field and potential is $\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$

$$\vec{E} = -[2y\hat{i} + (2x - z)\hat{j} + (-y)\hat{k}]$$

At point (1, 0, 1), the electric field is

$$\vec{E} = -(0\hat{i} + 1\hat{j} + 0\hat{k})$$

$$\vec{E} = -\hat{j}$$

(30) Answer : (3)

Solution:

Kirchhoff's loop law follows from conversation of energy and junction law follows from conservation of charge.

(31) Answer : (1)

Solution:

$$\phi = B.A \cos \theta$$

$$\phi = BA \cos \omega t$$

$$\varepsilon = -\frac{d\phi}{dt}$$

$$\varepsilon = BA\omega \sin \omega t$$

$$\frac{\sqrt{3}BA\omega}{2} = BA\omega \sin \omega t$$

$$\sin \omega t = \frac{\sqrt{3}}{2}$$

$$\omega t = \frac{\pi}{3}$$

(32) Answer : (3)

Solution:

$$i = \frac{V}{Z}$$

$$3 = \frac{300}{Z} \Rightarrow Z = 100 \Omega$$

$$Z = \sqrt{R^2 + (\omega L)^2}$$

$$(100)^2 = (60)^2 + (\omega L)^2 \Rightarrow L = 0.8 \text{ H}$$

(33) Answer : (1)

Solution:

Resistance doesn't vary with angular frequency.

$$\text{Capacitive reactance, } X_c = \frac{1}{\omega C}$$

$$\text{Inductive reactance, } X_L = \omega L$$

$$\text{Impedance } Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\text{At } \omega = \frac{1}{\sqrt{LC}}, Z_{\min} = R$$

(34) Answer : (2)

Solution:

$$\varepsilon_{\text{avg}} = \frac{\Delta \phi}{\Delta t}$$

$$= \frac{2\pi R^2 B}{\left(\frac{T}{2}\right)}$$

$$= 2R^2 B \omega$$

(35) Answer : (1)

Solution:

$$F = \frac{\mu_0 I_1 I_2}{2\pi r}$$

$$F \propto (i)^2$$

(36) Answer : (3)

Solution:

$$T = \frac{2\pi m}{qB}$$

$$\Rightarrow T \propto \frac{1}{B} \Rightarrow T \times B = C$$

$$\text{as } B \rightarrow 2B \rightarrow T \rightarrow \frac{T}{2}$$

(37) Answer : (3)

Solution:

A material suitable for making electromagnets should have low retentivity and low coercivity.

(38) Answer : (4)

Solution:

$$[\tau] = [ML^2T^{-2}]$$

$$[F] = [MLT^{-2}]$$

$\therefore \frac{\text{Torque}}{\text{Force}}$ is not dimensionless

$$\frac{\text{Torque}}{\text{Work}} = \frac{[MLT^{-2}]}{[MLT^{-2}]} = \text{Dimensionless}$$

Work and energy also have same dimension.

(39) Answer : (2)

Solution:

Magnitude of displacement is the shortest distance between initial and final position.

(40) Answer : (4)

Solution:

$$t \propto u$$

$$\frac{t_1}{t_2} = \frac{u_1}{u_2}$$

$$\frac{2}{t_2} = \frac{36}{144} \Rightarrow t_2 = 8 \text{ s}$$

(41) Answer : (3)

Solution:

Quantities that have different dimensions cannot be added.

(42) Answer : (1)

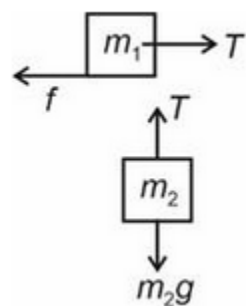
Solution:

$$K_1 = \frac{KL}{\frac{L}{3}} = 3K$$

$$K_2 = \frac{KL}{\frac{2L}{3}} = \frac{3K}{2}$$

(43) Answer : (2)

Solution:



$$T = f_{\max}$$

$$T = \mu N = 120 \text{ N}$$

$$T = m_2 g = 80 \text{ N}$$

$$\mu = \frac{2}{3}$$

(44) Answer : (2)

Solution:

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.4 \times \left(\sqrt{6^2 + (8)^2} \right)^2$$

$$= 20 \text{ J}$$

(45) Answer : (3)

Solution:

$$W = \int_0^{10} f_x dx$$

$$= \int_0^{10} (-0.6x^2 + 2.5) dx$$

$$= \left[-\frac{0.6x^3}{3} + 2.5x \right]_0^{10}$$

$$= |-0.2 \times 1000 + 2.5 \times 10|$$

$$W = -175 \text{ J}$$

$$W_f = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$-175 = \frac{1}{2} \times 2 (v_f^2 - 400)$$

$$400 - 175 = v_f^2$$

$$v_f = 15 \text{ m/s}$$

CHEMISTRY

(46) Answer : (4)

Solution:

(en) Ethylenediaminetetraacetate ion is a hexadentate ligand.

(47) Answer : (2)

Hint:

When negative charge on the metal increases, it can easily donate electrons to the vacant antibonding π^* orbital of carbon monoxide, it increases the M – C bond strength and decreases the C – O bond strength.

So, M – C bond strength: $[\text{Co}(\text{CO})_4]^- > [\text{Ni}(\text{CO})_4]$

C – O bond length: $[\text{Co}(\text{CO})_4]^- > [\text{Ni}(\text{CO})_4]$

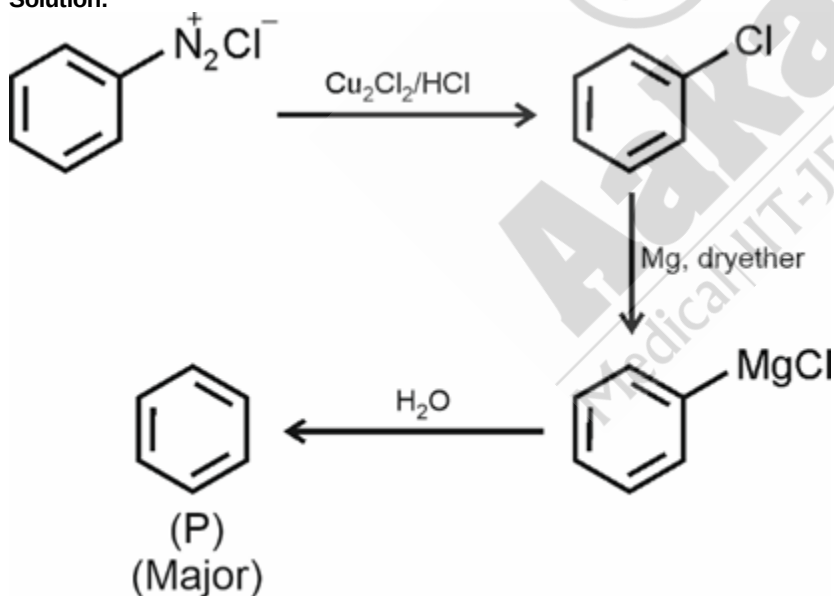
(48) Answer : (1)

Solution:

In haloarenes, C – X bond acquires a partial double bond character due to resonance. As a result, the bond cleavage in haloarene is difficult than haloalkane and therefore, they are less reactive towards nucleophilic substitution reaction.

(49) Answer : (4)

Solution:



(50) Answer : (1)

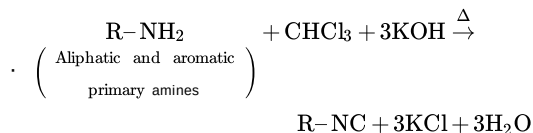
Solution:

Species	Boling point/K
$\text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$	310.5
$\text{C}_2\text{H}_5\text{CH}(\text{CH}_3)_2$	300.8
$\text{n-C}_4\text{H}_9\text{OH}$	390.3

(51) Answer : (3)

Solution:

- Only aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides or carbylamines which are foul smelling substances.
- Secondary and tertiary amines do not give this reaction.



(52) Answer : (3)

Solution:

a. Dry cell	–	Moist paste of NH_4Cl and ZnCl_2
b. Lead storage Battery	–	38% solution of H_2SO_4
c. Mercury cell	–	Paste of KOH & ZnO
d. $\text{H}_2 - \text{O}_2$ fuel cell	–	Concentrated aqueous NaOH solution

(53) Answer : (3)

Solution:

$$pH = pK_a + \log \frac{[\text{salt}]}{[\text{Acid}]}$$

$$6.76 = 4.76 + \log \frac{[\text{salt}]}{[\text{Acid}]}$$

$$2 = \log \frac{[\text{salt}]}{[\text{Acid}]}$$

$$10^2 = \log \frac{[\text{salt}]}{[\text{Acid}]}$$

$$\text{Salt : Acid} = 100 : 1$$

(54) Answer : (3)

Solution:

$$K_p = K_c (RT)^{\Delta n_g} \quad (\Delta n_g = 0)$$

$$K_p = 1.6 \times 10^5$$

(55) Answer : (4)

Solution:

$$n \times n_F = \frac{q}{F}$$

$$\frac{0.2}{40} \times 2 = \frac{9.65 \times t}{96500}$$

$$\frac{4}{400} = \frac{965 \times t}{965 \times 10^4}$$

$$\frac{10^4}{100} = t$$

$$t = 100 \text{ sec}$$

(56) Answer : (3)

Solution:

Eka – Aluminium is gallium
and Eka – silicon is germanium.

(57) Answer : (1)

Solution:

The elements after uranium are transuranium elements. The IUPAC nomenclature of atomic number 112 is Ununbium.

(58) Answer : (2)

Solution:

Higher the size of cation higher is the ionic character.

For cations of the same size and charge, the one with electronic configuration $(n-1)d^{1-10}ns^0$ has higher polarising power and covalent character. So ionic character $\text{ZnCl}_2 < \text{CaCl}_2$.

(59) Answer : (3)

Solution:

	Bond length
C – N	143 pm

C = N	138 pm
C $\equiv$ N	116 pm

(60) Answer : (1)

Solution:

Species	Bond angle
CH ₄	109.5°
NH ₃	107°
BF ₃	120°
H ₂ O	104.5°

(61) Answer : (1)

Solution:

$$P_A^\circ = 400 \text{ mm Hg} \quad P_B^\circ = 600 \text{ mm Hg}$$

$$P_{\text{total}} = P_A + P_B$$

$$= X_A P_A^\circ + X_B P_B^\circ$$

$$= X_A P_A^\circ + (1 - X_A) P_B^\circ$$

$$= X_A P_A^\circ + P_B^\circ - X_A P_B^\circ = P_B^\circ + (P_A^\circ - P_B^\circ) X_A$$

$$= 450 = 600 + (400 - 600) X_A$$

$$-150 = -200 X_A$$

$$X_A = 0.75$$

$$X_B = 0.25$$

(62) Answer : (4)

Solution:

$$\pi_{\text{NaCl}} = \pi_{\text{urea}}$$

$$iC_{\text{NaCl}} RT = iC_{\text{urea}} RT$$

$$2 \times C_{\text{NaCl}} = 0.2$$

$$C_{\text{NaCl}} = 0.1$$

(63) Answer : (4)

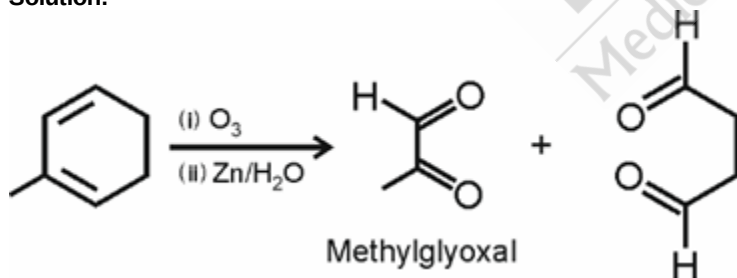
Solution:

For potassium, Z = 19

n, l, m_l and m_s for the valence electron of potassium atom are, 4, 0, 0 and $+\frac{1}{2}$ respectively.

(64) Answer : (3)

Solution:



(65) Answer : (1)

Solution:

Compounds	Structure
Thymine	→

Uracil	→	
Guanine	→	
Adenine	→	

(66) Answer : (4)

Solution:

Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff})

So correct order of energies of 2s orbital is

$H > Li > Na > K$

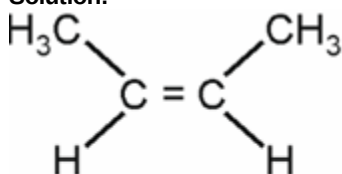
(67) Answer : (4)

Solution:

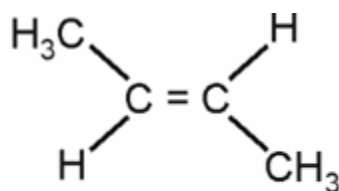
B group vitamins and vitamin C are soluble in water so they are grouped together.

(68) Answer : (2)

Solution:



cis-But-2-ene
($\mu = 0.33D$)

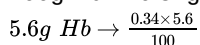
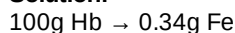


trans-But-2-ene
($\mu = 0$)

Melting point of trans-But-2-ene is higher than cis-But-2-ene because in trans-But-2-ene crystal packing is more efficient because of more symmetry.

(69) Answer : (2)

Solution:



$$\text{Fe(moles)} = 0.34 \times 10^{-3}$$

$$\text{Number of Fe atoms} = 0.34 \times 10^{-3} \times 6.02 \times 10^{23}$$

$$= 2.04 \times 10^{20}$$

(70) Answer : (4)

Solution:

$$\mu = \sqrt{n(n+2)} BM$$

$$\text{A. Ti}^{3+}; \mu = \sqrt{3} BM$$

$$\text{B. Sc}^{3+}; \mu = 0 BM$$

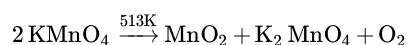
$$\text{C. Mn}^{4+}; \mu = \sqrt{15} BM$$

$$\text{D. Fe}^{3+}; \mu = \sqrt{35} BM$$

$$\text{E. Cr}^{3+}; \mu = \sqrt{15} BM$$

(71) Answer : (1)

Solution:



Potassium

permanganate

Potassium

Manganate

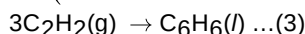
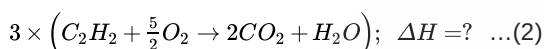
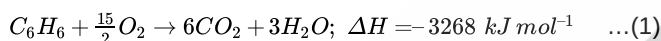
(72) Answer : (3)

Solution:

During boiling of an egg, denaturation of protein is responsible for increase in entropy of the system.

(73) Answer : (2)

Solution:



$$\Delta H = -632 \text{ kJ mol}^{-1}$$

$$\text{Eq. 3} = -(\text{eq. 1}) + 3 \text{ eq. 2}$$

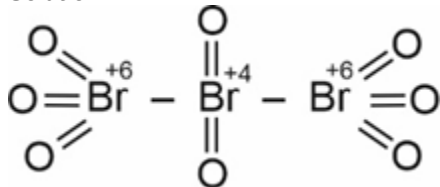
$$-632 = (3268) + 3x$$

$$-3900 \text{ kJ mol}^{-1} = 3x$$

$$-1300 \text{ kJ mol}^{-1} = x$$

(74) Answer : (3)

Solution:



(75) Answer : (1)

Solution:

Heavier members of the group 14 are more stable in the two unit less oxidation state than group oxidation state.

Hence, PbX_2 is more stable than PbX_4 while GeX_4 (Ge is in group oxidation state) is more stable than GeX_2 .

(76) Answer : (4)

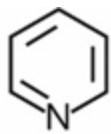
Solution:



Furan



Thiophene



Pyridine

Heterocyclic aromatic compounds



Tropone (Non-benzenoid, non-heterocyclic compound)

(77) Answer : (4)

Solution:

Actual pressure of nitrogen at 300 K temperature = $715 - 15 = 700$ mm

$$\text{Volume of N}_2 \text{ at STP} = \frac{273 \times 700 \times 60}{300 \times 760}$$

$$= 50.289 \approx 50.3 \text{ mL}$$

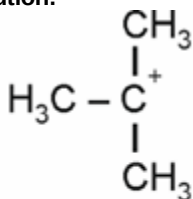
$$50.3 \text{ mL nitrogen weights} = \frac{28 \times 50.3}{22400} \text{ g}$$

$$\therefore \text{Percentage of nitrogen} = \frac{28 \times 50.3 \times 100}{22400 \times 0.28}$$

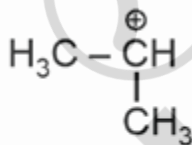
$$= 22.4\%$$

(78) Answer : (4)

Solution:

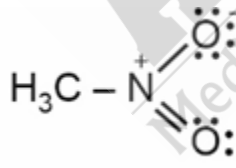
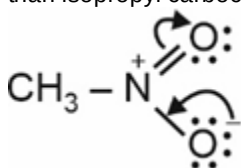


Tertbutyl carbocation
(9 α - Hydrogen)



Isopropyl carbocation
(6 α - Hydrogen)

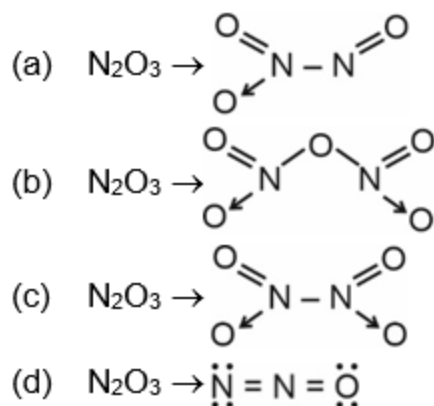
More hyperconjugation interaction (more α hydrogen) leads to more stability hence, tertbutyl carbocation is more stable than isopropyl carbocation



Two canonical forms of nitromethane

(79) Answer : (3)

Solution:



(80) Answer : (4)

Solution:

Chemically, red phosphorus is much less reactive than white phosphorus.

(81) Answer : (3)

Solution:

HOIO does not exist

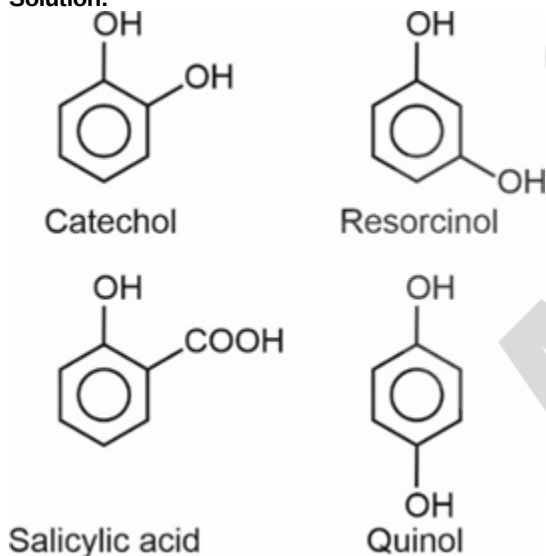
(82) Answer : (4)

Solution:

Tertiary alcohol is the most reactive towards Lucas reagent.

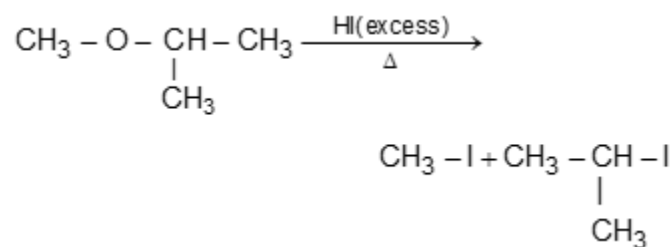
(83) Answer : (3)

Solution:



(84) Answer : (4)

Solution:



(85) Answer : (3)

Solution:

For 1st order reaction



$t_{1/2}$	$\frac{\ln 2}{k}$
$t_{75\%}$	$\frac{\ln 4}{k}$
$t_{99.9\%}$	$\frac{3 \ln 10}{k}$
$t_{90\%}$	$\frac{\ln 10}{k}$

(86) Answer : (2)

Solution:

For zero order reaction

$$[B] = kt = 2 \times 10^{-3} \times 2 \times 60 \\ = 0.24M$$

(87) Answer : (3)

Solution:

CH_3CH_2CHO and CH_3COCH_3 are functional isomer

(88) Answer : (1)

Solution:

Increasing order of reactivity towards nucleophilic addition reaction is
 $CH_3COCH_3 < C_6H_5CHO < CH_3CHO < HCHO$

(89) Answer : (4)

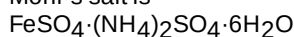
Solution:

Ba^{2+} imparts apple green colour during flame test

(90) Answer : (3)

Solution:

Mohr's salt is



BIOLOGY

(91) Answer : (2)

Solution:

Common names	Taxonomical categories
Mango	Dicotyledonae
Wheat	Poales
Brinjal	Polymoniales
Housefly	Muscidae

(92) Answer : (4)

Solution:

Archaeobacteria differ from other bacteria in having different cell wall structure but having same ribosomal subunits. All prokaryotes have 70S ribosomes composed of 30S and 50S subunits.

(93) Answer : (4)

Solution:

Toadstools are poisonous mushrooms which generally have white spores.

(94) Answer : (3)

Solution:

Zygomorphic flowers can be divided into two similar halves only in one particular vertical plane, e.g., *Cassia* and bean. Perianth condition is seen in lily, *Asparagus* etc.

(95) Answer : (2)

Solution:

Monodelphous – China rose

Epiphyllous – *Gloriosa*

Diadelphous – Pea

Polyadelphous – *Citrus*

(96) Answer : (4)

Solution:

- Prop root – Banyan tree
- Respiratory root – *Rhizophora*
- Fibrous root – Wheat plant
- Storage root – Sweet potato

(97) Answer : (2)

Solution:

Deuteromycetes is commonly known as imperfect fungi. The mycelium is septate and branched in them.

(98) Answer : (4)

Solution:

Krebs' cycle starts with the condensation of the acetyl CoA with oxaloacetic acid and water to yield citric acid.

(99) Answer : (2)

Solution:

Sugarcane is a C_4 plant. Primary CO_2 acceptor in sugarcane is PEP and it is a three carbon molecule.

(100) Answer : (2)

Solution:

Species composition and stratification are the structural components of an ecosystem. Earthworm, an eukaryotic organism breaks the detritus into the smaller particles by the process known as fragmentation.

(101) Answer : (4)

Solution:

The diagram is of C_4 pathway, in which P represents fixation, Q represents decarboxylation and R represents the regeneration process.

(102) Answer : (4)

Solution:

In the process of glycolysis, substrate level phosphorylation reaction occurs at two steps. Net gain of ATP is 2 ATP molecules. Release of water occurs during conversion of 2-phosphoglycerate to phosphoenolpyruvate.

(103) Answer : (2)

Solution:

Of the incident solar radiation, less than 50% of it is photosynthetically active radiation. Plants capture only 2-10% of PAR or 1-5% of incident solar radiation for synthesis of organic matter.

(104) Answer : (2)

Solution:

Formation of interfascicular cambium is an example of dedifferentiation.

(105) Answer : (2)

Solution:

Coriander and cotton show developmental heterophylly. Cytokinins are widely applied in tea plantations. Kinetin was first identified and crystallised by Miller et al.

(106) Answer : (1)

Solution:

Calcium plays a very important role in blood clotting and it constitutes 1.5% weight of the human body and 3.6% weight of Earth's crust.

(107) Answer : (3)

Solution:

The chemical and physical properties of amino acids are essentially due to the amino, carboxyl and the R functional groups. Based on the number of amino and carboxyl groups, there are acidic (e.g., glutamic acid), basic (lysine) and neutral (valine) amino acids. Similarly, there are aromatic amino acids (tyrosine, phenylalanine, tryptophan).

(108) Answer : (2)

Solution:

Palmitic acid has 16 carbons including carboxyl carbon. Arachidonic acid has 20 carbon atoms including the carboxyl carbon. Palmitic acid is a saturated fatty acid while arachidonic acid is an unsaturated fatty acid possessing $C = C$ bonds in its structure.

(109) Answer : (4)

Solution:

Primary metabolites have identifiable functions and play known roles in normal physiological processes. We do not at the moment, understand the role or functions of all the 'secondary metabolites' in host organisms. However, many of them are useful to 'human welfare' (e.g., rubber, drugs, spices, scents and pigments).

(110) Answer : (1)

Solution:

If the columnar or cuboidal cells bear cilia on their free surface, they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

(111) Answer : (4)

Solution:

The movement of air into and out of the lungs occurs due to pressure gradient.

(112) Answer : (2)

Solution:

The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking. Association areas are responsible for complex functions like intersensory associations, memory and communication.

The cerebral hemispheres are connected by a tract of nerve fibres called corpus callosum.

The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions.

(113) Answer : (4)

Solution:

The first isolated restriction endonuclease is *Hind* II. It was found that *Hind* II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs.

(114) Answer : (2)

Solution:

Micro-injection is a direct method of gene transfer used for animals, whereas retrovirus mediated is an indirect method of gene transfer in animals.

In biolistics or gene gun, cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA. This is a direct method of gene transfer and is suitable for plants.

Ti-plasmid mediated transfer is an indirect method of gene transfer for plants.

(115) Answer : (4)

Solution:

DNA ligase facilitates the joining of sticky ends obtained after restriction digestion of a desired DNA. It belongs to class VI of enzymes.

(116) Answer : (4)

Solution:

The proximal end of the filament of stamen is attached to the thalamus or the petal of a flower.

(117) Answer : (3)

Solution:

In the given figure, A represents scutellum.

(118) Answer : (3)

Solution:

Perisperm is found in beet. Groundnut seeds are non-endospermic as the endosperm gets completely consumed by the developing embryo. *Salvia* is an insect pollinated plant.

(119) Answer : (2)

Solution:

Mendel found that the phenotype of the F_1 heterozygote (Tt) was exactly like the parent with dominant genotype (TT), i.e. both were tall.

(120) Answer : (3)

Solution:

Sickle-cell anaemia is controlled by a single pair of allele, Hb^A and Hb^S . β -thalassemia is controlled by a single gene HBB on chromosome 11 of each parent.

(121) Answer : (1)

Solution:

Skin colour in humans – Polygenic inheritance

XO type of sex determination – Grasshopper

Female heterogamety – Birds

Phenylketonuria – Pleiotropy

(122) Answer : (2)

Solution:

	I^A	I^B

i	A _i	B _i
---	----------------	----------------

So the possible blood types of the offspring are only A and B.

(123) Answer : (2)

Solution:

The above pedigree represents an autosomal recessive trait, an example of which is sickle-cell anaemia.

(124) Answer : (2)

Solution:

Intrafascicular vascular cambium and cork-cambium are examples of lateral meristems.

(125) Answer : (4)

Solution:

In monocot stem, vascular bundles are found scattered in whole ground tissue. The larger vascular bundles are loosely arranged towards the centre of the stem.

(126) Answer : (2)

Solution:

Guanosine is a nucleoside, in which a nitrogenous base is linked to sugar through a N- glycosidic linkage. When phosphate group is linked to nucleoside through phosphoester bond, a corresponding nucleotide is formed.

(127) Answer : (4)

Solution:

The first phase of translation is aminoacylation of tRNA.

(128) Answer : (3)

Solution:

AUG codon is a non-degenerate codon. AUG codes for methionine only.

(129) Answer : (2)

Solution:

Guard cells have thin outer walls and thick inner walls. In monocot leaf, hypodermis is absent. The radial conduction of water takes place by the ray parenchymatous cells.

(130) Answer : (3)

Solution:

The length of DNA for human diploid cell is $6.6 \times 10^9 \text{ bp} \times 0.34 \times 10^{-9} \text{ m} = 2.2 \text{ metres}$.

(131) Answer : (4)

Solution:

Recombinant DNA technology involves several steps in specific sequence such as isolation of DNA, fragmentation of DNA by restriction endonucleases, isolation of a desired DNA fragment, ligation of the DNA fragment into a vector, transferring the recombinant DNA into the host (transformation), culturing the host cells in a medium at large scale and extraction of the desired product called downstream processing.

(132) Answer : (4)

Solution:

Nissl's granules are made up of rough endoplasmic reticulum (RER) and free ribosomes, and are found in the dendrites and cyton of neurons. They are not found in axons.

(133) Answer : (3)

Solution:

The number of DNA fragments obtained after the restriction digestion of a circular DNA is equal to the number of restriction sites present for a particular restriction endonuclease

In the given question, the circular DNA has four and two restriction sites for *Pst* I and *Pvu* I respectively, but since we are performing the restriction digestion with the enzymes *Pvu* II and *Pst* I, thus only four DNA fragments will be obtained as *Pvu* II will not be able to show its action as there is no restriction site for this enzyme. Only *Pst* I will act to generate four restriction fragments.

(134) Answer : (3)

Solution:

Sternum is the part of axial skeleton.

(135) Answer : (3)

Solution:

The P-wave represents the electrical excitation (or depolarisation) of the atria, which leads to the contraction of both the atria. The QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction. The contraction starts shortly after Q and marks the beginning of the systole. The T-wave represents the return of the ventricles from excited to normal state (repolarisation). The end of the T-wave marks the end of ventricular systole.

(136) Answer : (4)

Solution:

During muscle contraction, only actin filaments are pulled towards centre of the 'A' band from the components of myofilament.

(137) Answer : (2)

Solution:

Stegosaurus was plated back and a herbivore animal.

Brachiosaurus was long necked herbivore.

Pteranodon was a flying reptile.

Triceratops had a three horned face.

(138) Answer : (4)

Solution:

During strenuous workout, respiratory rate, heart rate, strength of ventricular contraction and cardiac output increases.

Parasympathetic neural signals decrease the rate of heart beat, speed of conduction of action potentials and thereby the cardiac output.

(139) Answer : (1)

Solution:

Homo habilis – 650 cc - 800 cc

Neanderthal man – 1400 cc

Homo erectus – 900 cc

(140) Answer : (4)

Solution:

The seminiferous tubules of the testis open into the vasa efferentia through rete testis.

(141) Answer : (1)

Solution:

Placenta is the structural and functional link between developing embryo and maternal body.

(142) Answer : (4)

Solution:

Sensory papillae, taste buds and nasal epithelium are cellular aggregations around nerve endings but eyes and internal ears are well organised structures in frogs.

(143) Answer : (2)

Solution:

The epididymis leads to vas deferens that ascends to the abdomen and loops over the urinary bladder that receives ureter from the kidneys and opens into urethra. Ejaculatory duct is formed by union of duct of seminal vesicle and vas deferens and open into urethra.

(144) Answer : (4)

Solution:

In male cockroaches, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by the 9th sternum. It contains dorsal anus, ventral male genital pore and gonapophyses.

(145) Answer : (4)

Solution:

A functional mammary gland is characteristic of all female mammals. Oogenesis in females begins during embryonic development whereas spermatogenesis begins at puberty in males.

(146) Answer : (1)

Solution:

Nicotine is a stimulant and an alkaloid.

Tobacco contains a large number of chemical substances including nicotine, an alkaloid.

Nicotine stimulates adrenal gland (adrenal medulla) to release adrenaline and nor-adrenaline into blood circulation both of which raise blood pressure and increase heart rate.

(147) Answer : (2)

Solution:

Hormones are produced by endocrine glands and released into the blood and transported to a distantly located target organs. Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts. They are secreted by organised endocrine glands and diffused tissues located in different parts of the body.

(148) Answer : (3)

Solution:

The exaggerated response of the immune system to certain antigens present in the environment is called allergy. Such substances to which such an immune response is produced are called allergens. The antibodies produced to these are IgE type. Common examples of allergens are mites in dust, pollens, animals dander etc.

(149) Answer : (1)

Solution:

Exophthalmic goitre is a form of hyperthyroidism, characterised by enlargement of thyroid gland, protrusion of eyeballs, increased BMR and weight loss, also called Graves' disease.

- Gigantism is due to over-secretion of growth hormone.
 - Addison's disease is due to underproduction of adrenal cortex hormones.
- Prolonged hyperglycaemia leads to diabetes mellitus.

(150) Answer : (3)

Solution:

X-rays and gamma rays are ionising radiations while UV rays are non-ionizing radiations. Both are physical carcinogens capable of transforming normal cells into cancerous cells.

(151) Answer : (4)

Solution:

Diseases can be infectious and non-infectious. For eg. AIDS, polio, diphtheria, tetanus, etc are infectious diseases. Among non-infectious disease, cancer is one of the major cause of death.

(152) Answer : (3)

Solution:

Cyclostomes possess cartilaginous cranium and vertebral column. They lack scales and paired fins but possess unpaired fins. *Myxine* belongs to the class Cyclostomata. *Scoliodon* is a cartilaginous fish and possess placoid scales. *Betta* and *Pterophyllum* are bony fishes.

(153) Answer : (3)

Solution:

Darwin's finches show adaptive radiation, homology, divergent evolution, founder effect and natural selection. Analogy is the result of convergent evolution.

(154) Answer : (2)

Solution:

Evolution is not a directed process in the sense of determinism. Bandicoot and wombat show adaptive radiation/divergent evolution. Sweet potato (root modification) and potato (stem modification) is example of analogy.

(155) Answer : (3)

Solution:

PCT helps to maintain the pH and ionic balance of the body fluids by selective secretion of H^+ and ammonia into the filtrate and by absorption of HCO_3^- from it.

(156) Answer : (4)

Solution:

In eukaryotes, regulation could be exerted at

- (i) Transcriptional level
- (ii) Processing level
- (iii) Translational level
- (iv) Transport of mRNA from nucleus to cytoplasm

(157) Answer : (3)

Solution:

Gas vacuole are membraneless vacuoles, found in prokaryotes, which provide buoyancy.

(158) Answer : (4)

Solution:

- (i) B-Rough endoplasmic reticulum helps in protein synthesis.
- (ii) C-Ribosomes are composed of ribonucleic acid and proteins and not surrounded by any membrane.

(159) Answer : (2)

Solution:

(i)	Nucleolus	–	Site for ribosomal RNA synthesis
(ii)	Cristae	–	Infolding of mitochondrial inner membrane
(iii)	Cisternae	–	Flattened sac-like

				structures stacked on one another
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(160) Answer : (4)**Solution:**

During the pachytene stage, the four chromatids become distinct and clearly appear as a tetrad.

(161) Answer : (1)**Hint:**

G₀ is the quiescent stage of cell cycle, as cells in this phase are metabolically active but no longer proliferate.

(162) Answer : (4)**Solution:**

Astral rays are made up of microtubules, which arise from the centrosomes.

(163) Answer : (2)**Solution:**

Gemmae are green multicellular, asexual buds, which develop in gemma cups. The predominant stage of the life cycle of a moss is the gametophyte which consists of two stages, the first stage is protonema and second stage is leafy stage.

(164) Answer : (3)**Solution:**

Salvinia – Heterosporous pteridophyte

(165) Answer : (3)**Solution:**

Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment.

(166) Answer : (3)**Solution:**

Some snails and fish go into aestivation to avoid summer related problems.

(167) Answer : (4)**Solution:**

In co-extinction, when a species extinct, the plant and animal species associated with it, in an obligatory way, also becomes extinct.

(168) Answer : (4)**Solution:**

Wildlife sanctuaries and zoological parks are examples of *in-situ* conservation and ex-situ conservation, respectively.

(169) Answer : (3)**Solution:**

Citric acid is commercially produced from *Aspergillus niger*.

(170) Answer : (2)**Solution:**

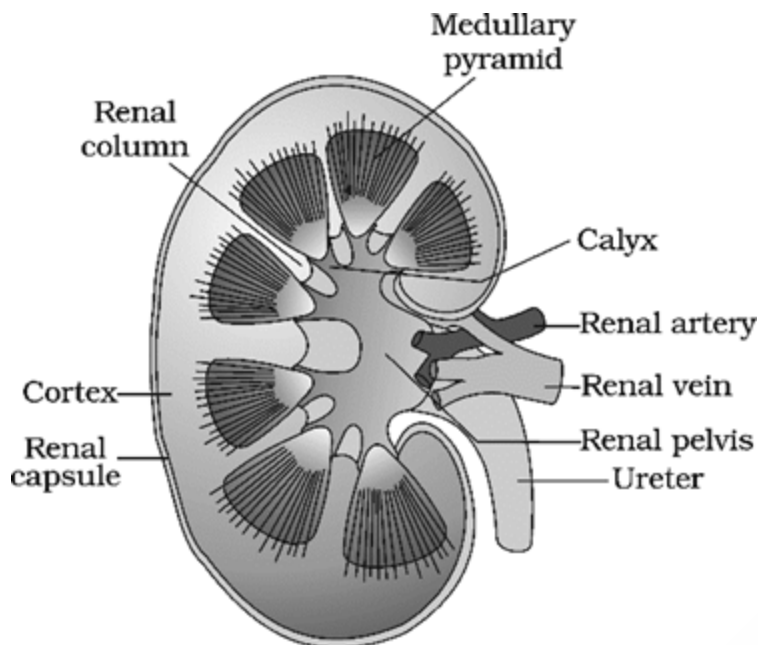
The interaction between sea anemone and clown fish is commensalism. The fish gets protection from predators which stay away from the stinging tentacles of sea anemone. The anemone does not appear to derive any benefit by hosting the clown fish.

(171) Answer : (4)**Solution:**

Skeletal muscle fibres are striated, cylindrical shaped and multinucleated with peripheral nuclei. Urinary bladder has smooth muscle fibres which are uninucleated and fusiform shaped. Both of them are unbranched.

(172) Answer : (4)**Solution:**

'D' is ureter which carries filtrate (urine) which may be four times concentrated than the initial filtrate formed.



(173) Answer : (3)

Solution:

Arthropods have open circulatory system and three germ layers. Insects have Malpighian tubules as their excretory structures, green glands present in crustaceans for excretion. Arthropods are segmented animals.

(174) Answer : (3)

Solution:

Bones have a hard and non-pliable ground substance

(175) Answer : (2)

Solution:

Chiton – Unsegmented body

Filarial worm – Body cavity not lined by mesoderm

Locust – Terrestrial insect

Squid – Organ system level of organisation

(176) Answer : (3)

Solution:

Chain A and chain B are connected by disulphide bonds in mature human insulin.

(177) Answer : (1)

Solution:

Blood is the medium of transport for O_2 and CO_2 . About 97 per cent of O_2 is transported by RBCs in the blood. The remaining 3 per cent of O_2 is carried in a dissolved state through the plasma. Nearly 20-25 per cent of CO_2 is transported by RBCs whereas 70 per cent of it is carried as bicarbonate. About 7 per cent of CO_2 is carried in a dissolved state through plasma.

(178) Answer : (4)

Solution:

Lactational amenorrhea (absence of menstruation) contraception method is based on the fact that ovulation and therefore the menstrual cycle do not occur during the period of intense lactation following parturition.

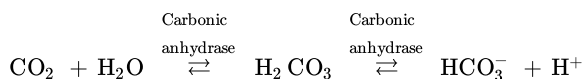
Administration of progestogens or progestogen-estrogen combinations or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or casual unprotected intercourse.

Saheli is a non-steroidal oral contraceptive pill.

(179) Answer : (3)

Solution:

RBCs contain a very high concentration of the enzyme, carbonic anhydrase and minute quantities of the same is present in the plasma too.



At the tissue site where partial pressure of CO_2 is high due to catabolism, CO_2 diffuses into blood (RBCs and plasma) and forms HCO_3^- and H^+ . At the alveolar site where pCO_2 is low, the reaction proceeds in the opposite direction leading to the formation of CO_2 and H_2O .

(180) Answer : (4)

Solution:

One could be free of these infections by following the simple principles given below: (i) Avoid sex with unknown partners/multiple partners. (ii) Always try to use condoms during coitus. (iii) In case of doubt, one should go to a qualified doctor for early detection and get complete treatment if diagnosed with infection.

